Reproducible Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Reproducible Experiments before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Reproducible Experiments trace and record the first step you can explain without help.

Use W2: Improve a record for Experimental records and **seed**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Reproducible Experiments explanation against the supplied facts before discussing it.

Use the worked case for Experimental records and **seed**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Reproducible Experiments target to a trace and a justified conclusion.

Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Reproducible Experiments case and show the deciding evidence.

A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.

Specify **reproducibility** records.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 26 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Reproducible Experiments adds to Responsible Plant Classifier and keep the evidence.

Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, **prediction**s, and metrics.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Summarize all runs

> Summarize all runs. Apply the supplied Reproducible Experiments facts and keep your reasoning.

Calculate the worked mean and range in percentage points.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Improve a record

> Improve a record. Apply the supplied Reproducible Experiments facts and keep your

> reasoning.

A file contains only **seed** 22 and 80 percent. Name four missing items.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Separate two ideas

> Separate two ideas. Apply the supplied Reproducible Experiments facts and keep your reasoning.

Does identical **output** prove a valid experiment?

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Reproducible Experiments facts and keep your reasoning.

Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

My reasoning and evidence

W5 Handle a failed run

> Handle a failed run. Apply the supplied Reproducible Experiments facts and keep your reasoning.

One planned **seed** fails before **prediction**. How should it appear in the record?

My reasoning and evidence

WORKBOOK · 17 OF 26

Versioned experiments — W1 Record a run

> Use the supplied Reproducible Experiments task to record your reasoning, evidence and next step.

List five items needed to reproduce a scoring run.

My reasoning and evidence

WORKBOOK · 18 OF 26

Versioned experiments — W2 Compute a comparison

> Use the supplied Reproducible Experiments task to record your reasoning, evidence and next step.

A has fun 3 and learning 2. Score it under weights (2,1) and (1,2).

My reasoning and evidence

WORKBOOK · 19 OF 26

Versioned experiments — W3 Repair an experiment

> Use the supplied Reproducible Experiments task to record your reasoning, evidence and next step.

A learner changes weights, **input** ratings and the time limit together. Why is attribution difficult?

My reasoning and evidence

WORKBOOK · 20 OF 26

Versioned experiments — W4 Design independently

> Use the supplied Reproducible Experiments task to record your reasoning, evidence and next step.

Plan two **version**s for C (fun 5, learning 1) and D (fun 2, learning 5), using weights (2,1) and (1,2). Record expected scores, ranks, fixed data and a separate final-case plan.

My reasoning and evidence

WORKBOOK · 21 OF 26

Versioned experiments — W5 Protect evaluation

Use the supplied Reproducible Experiments task to record your reasoning, evidence and next step.

You revise the policy after reading final-case failures. Are those cases still untouched **holdout evidence**?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Reproducible Experiments

> Record variation across declared seeds instead of selecting the prettiest run. Explain the labelled step before checking the changed condition.

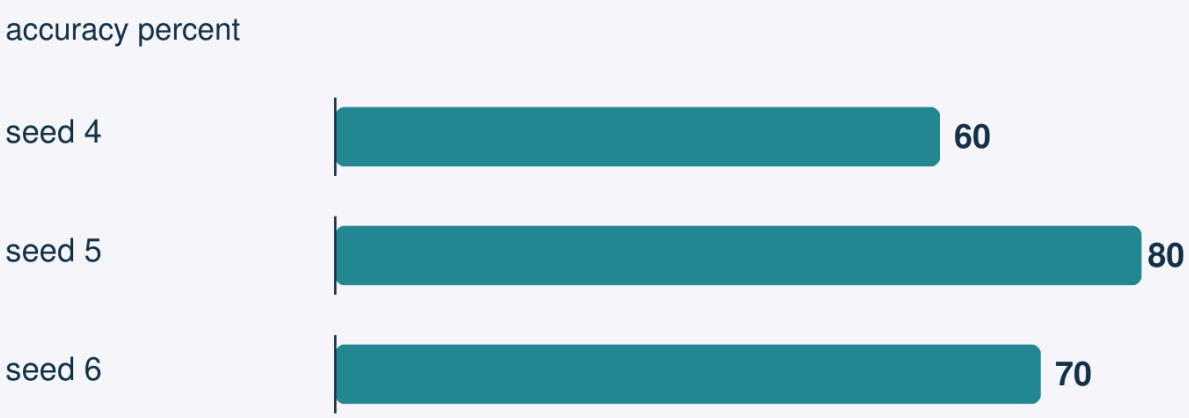
Record variation across declared **seed**s instead of selecting the prettiest run.

Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/26

All declared seeds



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/26

Reproducible Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Keep run identities

Record seeds4, 5, 6 with 6/10, 8/10, 7/10.

2. Summarise variation

Mean accuracy is 70%; range is 60% to80%.

3. Preserve context

Record data version, split, code, settings and seeds, not just the winning80%.

CHANGE ONE THING

The report deletes the 60% run.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Record variation across declared **seed**s instead of selecting the prettiest run.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.

1. Preserve context

2. Keep run identities

3. Summarise variation

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Preserve context Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Read the picture in order.

Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The report deletes the **60%** run. Which conclusion is supported by the supplied facts?

1. The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Repair a missing building block

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised **version**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

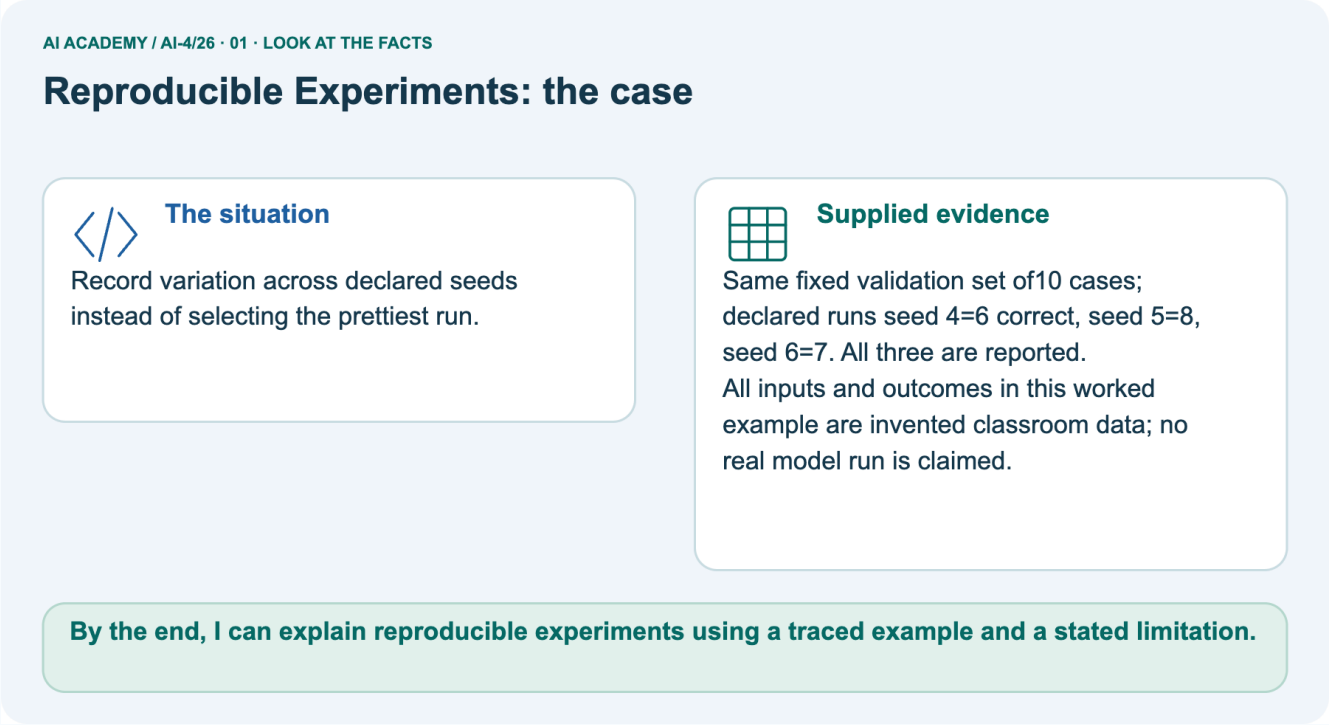
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Reproducible Experiments picture from its labels, then record a relevant limit.

01. Reproducible Experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

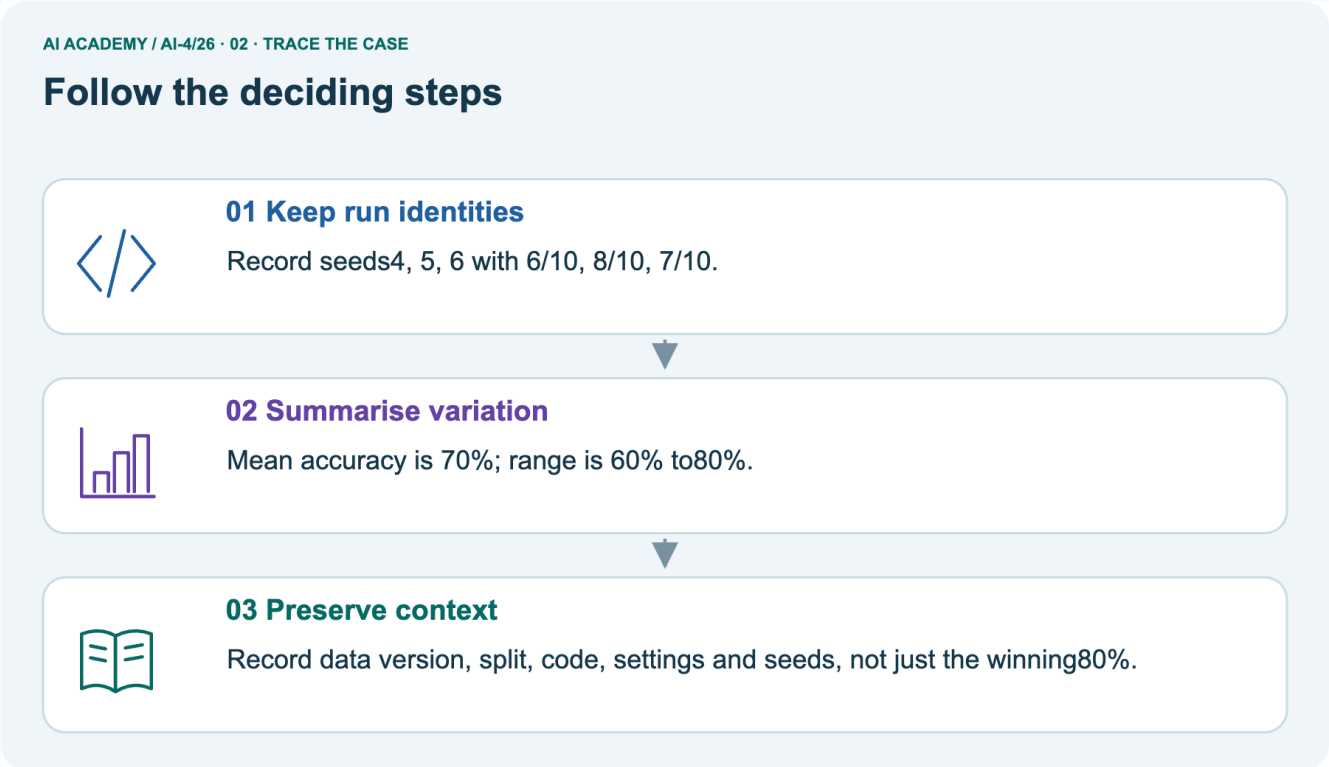
Record variation across declared **seed**s instead of selecting the prettiest run. Supplied facts: Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep run identities: Record **seed**s4, 5, 6 with 6/10, 8/10, 7/10. → Summarise variation: Mean accuracy is **70%**; range is **60%** to**80%**. → Preserve context: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. All declared seeds



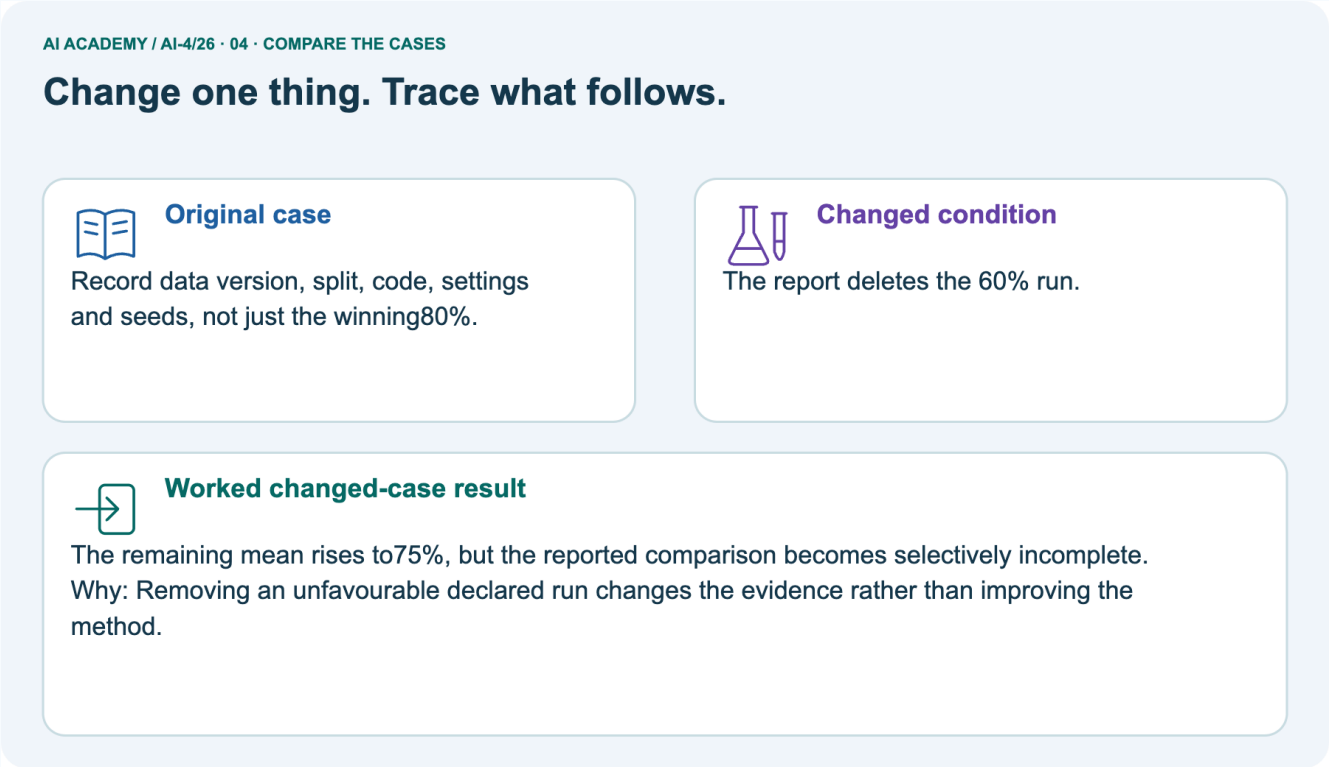
Follow the labelled picture with the matching explanation. Open the vector diagram

All declared **seed**s. Units: accuracy percent. **seed** 4: 60. **seed** 5: 80. **seed** 6: 70.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

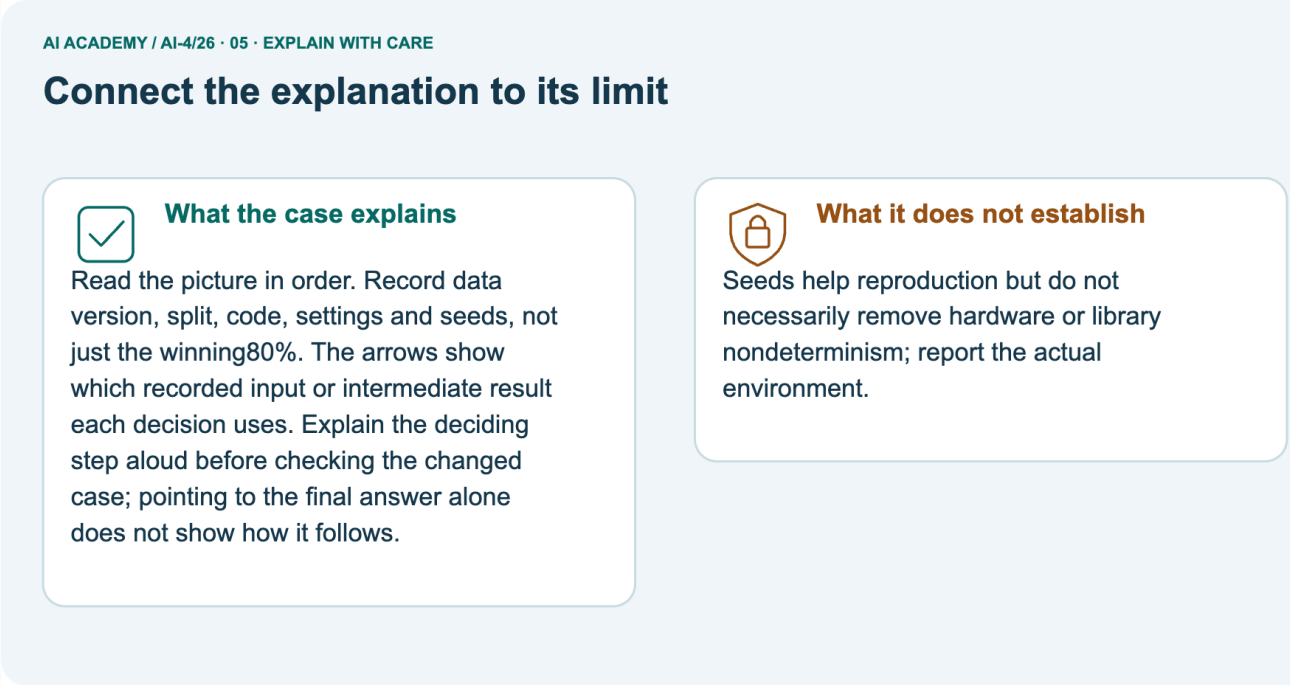
Original case: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.
Changed condition: The report deletes the **60%** run.

Expected result: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Seed**s help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

> Explain the case, then point to a limit on the claim.

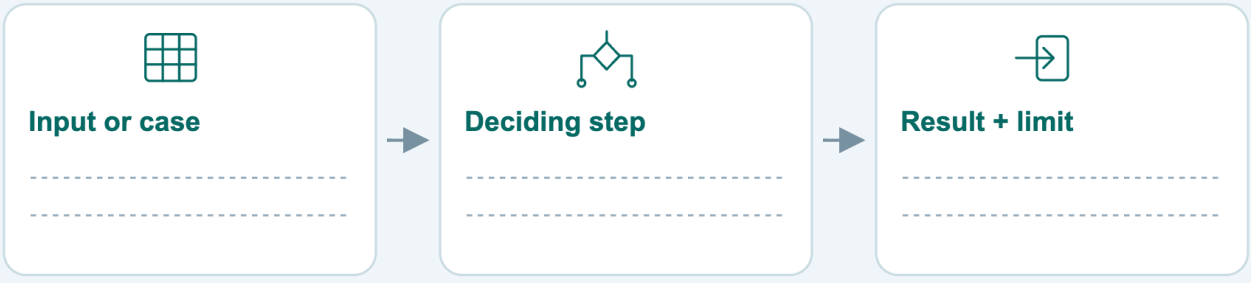
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-4/26 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.